# Research Statement

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My research is focused on applying sampling-based motion planners to constrained systems. Many constraints take the form of dynamic constraints, or constraints on the direction, velocity, and acceleration. Planning under dynamic constraints (i.e. kinodynamic planning) has applications in areas such as mobile robots, aerial robotics, and dexterous manipulation. Kinodynamic planning is especially difficult because there is no solution to the two-point boundary problem, which means that we must rely solely on expansion through forward propagation, making methods such as the PRM[1] and RRT*[2], which require a local planner, unsuitable. Constraints could also take the form of end-effector constraints or constraints introduced by closed chains. Such constraints are common to manipulation tasks found in areas such as industrial automation, and are difficult because the constraints take the form of a manifold in planning space, and traditional sampling-based methods cannot be applied because of the difficulty associated with generating samples on manifolds.

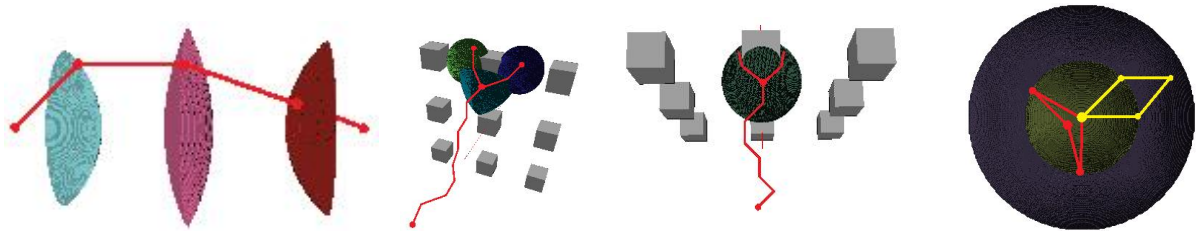


Figure 1: **Grasping on the move in kitting environment**

My research leverages machine learning and geometric solutions to do motion planning in constrained systems. I apply learning as a surrogate for aspects of the problem that are either infeasible or computationally efficient, such as a solution to the two-point boundary problem in kinodynamic planning and as a heuristic to help guide planning. I also use geometric models to develop visualizations of problems in order to better understand them, and I apply these models to develop planning spaces in which we can use sampling-based planners to produce constraint-satisfying motions.

## Motion Planning for Manipulations and Constrained Systems:

Motion planning with constraints has applications in parallel robotics [3], grasping and manipulation [4], computational biology and molecular simulations [5], and animation [6]. Constrained motion planning problems place constraints on the motion of an object (robot). These constraints might require that the robot remain in contact with a surface or maintain a specific clearance. They could also require that certain joints of the robot remain in contact with each other (e.g., closed chains). Such constraints could be used to constrain the graspers of a manipulator to a set of grasping positions, or handles. They could also be used in industrial automation to constrain a tool mounted on a robot to a surface or a seam (for example, we could constrain a welder mounted on a robot to a seam which needs to be welded). Constraints could also be used to simulate contacts or bindings in protein folding simulations.



(a) (b) (c) (d)

Figure 2: **The reachable volumes (a) of a 4 link chain with spherical joints with its end effector constrained to a point. The reachable volumes of a 16 degree of freedom fixed-base grasper with spherical joints is affected by constraints placed either (b) on the end effectors to be within spherical regions of the object or (c) on the base to be at a point. Note that in (c) the end effector reachable volumes are identical so only one is shown. The reachable volume (d) of a closed chain with 4 spherical joints and 4 links of equal length. The first and third joints can reach any point along the inner sphere (green) while the second joint can reach any point inside the outermost sphere (blue). Example configurations shown in red and yellow (d).**

Sampling-based motion planning methods such as the graph-based PRM [1] and the tree-based RRT [7] are state of the art solutions to traditional motion planning problems. Unfortunately, these methods are poorly suited for many constrained problems where the constraints form a manifold in C-space and planning must be restricted to this manifold [8]. Previous methods have developed specialized samplers that generate samples that satisfy constraints [9, 10, 11], these methods are either unable to handle high degree of freedom systems or unsuited for systems with spherical or prismatic joints or systems that combine different types of joints. These methods only guarantee that samples lie on the constraint, while my methods ensure that local paths between samples also satisfy constraints, resulting in constraint-satisfying trajectories.

**Motion Planning with Reachable Volumes:** The primary focus of my thesis work is motion planning for constrained systems and high degree of freedom robots. I introduce the concept of *Reachable Volumes (RV)* [12, 13], which are a geometric representation of the regions the joints and end effectors of a robot can reach, and use it to define a new planning space called RV-space where all points automatically satisfy a problem's constraints. Samples and paths generated in RV-space naturally conform to constraints, making planning for constrained systems no more difficult than planning for unconstrained systems. Consequently, constrained motion planning problems that were previously difficult or unsolvable become manageable and in many cases trivial.

Visualizations of RVs have applications in robot control and operation, where they can assist an operator by showing them what regions their robot can reach from a given location. This will help them decide where and how to position the robot to accomplish their goal. For example, they could show the operator of a grasper where they need to position their robot in order to grasp and manipulate an object. They can also assist in robot design by indicating whether the robot can reach parts of the environment that it needs to in order to accomplish its task. Figures 4 and 2 show examples of RVs for a variety of robots.

My work introduces tools and techniques to extend the state of the art sampling-based motion planning algorithms to RV-space. I propose an RV sampler, an RV local planner, and an RV distance metric. RV sampling generates samples by iteratively sampling the joints of a robot in their RV, resulting in samples that are guaranteed to satisfy a problems constraints. It can solve problems with constraints applied to any combination of joints/end-effectors, while most other methods (e.g. [14, 11, 15]) assume a single constraint, usually on one of the end-effectors. The RV local planner and distance metric can be used to generate constraint-satisfying local paths, even in problems such as closed chains where the constraints

form a manifold. As part of the RV local planner, I present a novel method for stepping RV samples to generate samples that are close to the original while ensuring that constraints are satisfied.

The geometric complexity of RVs is $\mathcal{O}(1)$ in unconstrained problems and in many constrained problems. This allows us to generate samples in linear time with respect to the number of bodies in the robot, which is the best possible complexity for a sampler. In problems with more complex constraints, samples can be generated in $\mathcal{O}(|L|^2\ |S|$ Complexity$(S))$ time, where $S$ is the set of constraints
and $|L|$ is the number of bodies in the robot. I also show that RV planners are probabilistically complete.

## Grasping on the Move:

My current work combines motion planning and visual servoing to do grasping on the move for mobile manipulators. Grasping on the move has a wide range of applications ranging from industrial automation to home robotics, to search and rescue, and space exploration. Solutions must achieve high precision in order to achieve a successful grasp, and they must achieve precise timing to intercept and grasp the object as the robot passes. They must also address any noise and uncertainty caused by the motion of the base and adapt to any changes in the environment. Furthermore, planning time should be minimized to ensure efficient operation.

I propose an architecture for grasping on the move in order to facilitate pick and place operations while a mobile manipulator's base is in motion. My architecture eliminates the need for the robot to stop and localize in order to perform a pick and place operation which improves efficiency in settings such as industrial automation.

This framework includes planners and controllers for the base and arm, as well as an object detector. The base planner/controller uses LiDAR sensors to navigate the work environment in order to execute paths that pass within reachable distance of objects that need to be grasped. A vision-based object detector identifies and localizes grasp objects using a camera mounted on the arm. The arm planner/controller incorporates base motion and object location to grasp objects detected by the vision system.

I show application to a mobile manipulator robot, which consists of a UR5 arm mounted on a MiR mobile base, and demonstrate my method on kitting tasks in an industrial environment. I also show that it can consistently identify and grasp target objects while the robot's base is undergoing dynamic motion.

## Application of Machine Learning to Kinodynamic Planning for Terrain Environments:

Motion planning for off-road vehicles is a difficult open problem due to the many factors that influence robot motion. In addition to the dynamics of the system, a planner must take into account the topology of the environment, the driving characteristics of the terrain (e.g. traction, friction, etc) and the presence of impassable regions where the robot could become stuck. Furthermore, terrain environments may be extremely large (the Gravel Pit environment shown in Figure 3 consists of a 1.1 km circuit). Real-world environments may also be unmapped, or not mapped in detail, which means that a motion planner needs the ability to plan and re-plan based on the robot's local observations.

My research focuses on using learning to intelligently select controls to apply during forward propagation. In particular, I develop a learned model that is used as a surrogate for a steering function, which generates controls that approximately reach a specified goal. This controller is parameterized on the robot's state and the topology of the terrain at the robot's position. I train this controller in a set of sloped environ-

ments using the Soft Actor-Critic (SAC) architecture[16] with Hindsight Experience Replay (HER) [17]. A variation of this learned model can also be used as[16]a cost-to-go.

One major advantage of this controller is that it only requires local knowledge of the environment in the form of a tangent vector at the robot's current position. In real-world robots this information can be obtained from on-board sensors or from performing PCA on point-cloud representation of the local terrain. This locality makes the terrain-aware controller well suited to real-world applications where the robot may only know the local topology of the environment due to limited sensor range. This representation is also advantageous because it reduces the terrain to three parameters, which limits the complexity of the controller. I show experimentally that sampling-based planners using this controller can generate paths for mobile robots in a wide variety of complex terrains, including the gravel pit shown in figure 3. My results demonstrate that the terrain-aware controller significantly improves planner performance and yields better quality (lower cost) paths than previous methods.

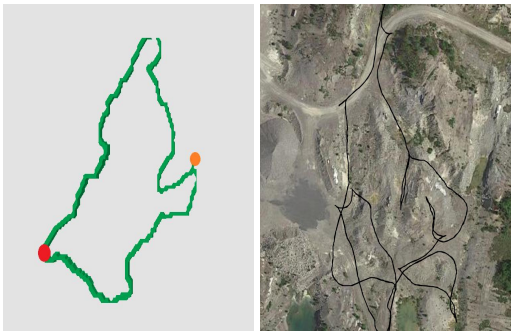
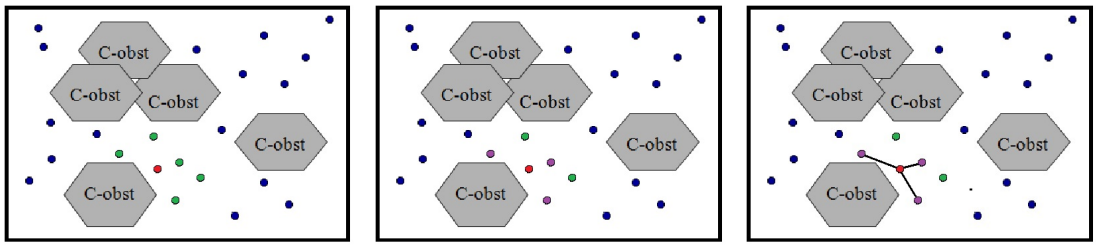


Figure 3: (top-left) Experimental environment constructed using the *Gravel Pit Lidar-Intensity Imagery Dataset*[18], which is generated from LIDAR scans of a 1.1 km of a circuit. (top left) Satellite image of the terrain based on which the dataset is generated [18].

## Addressing motion planning fundamentals:

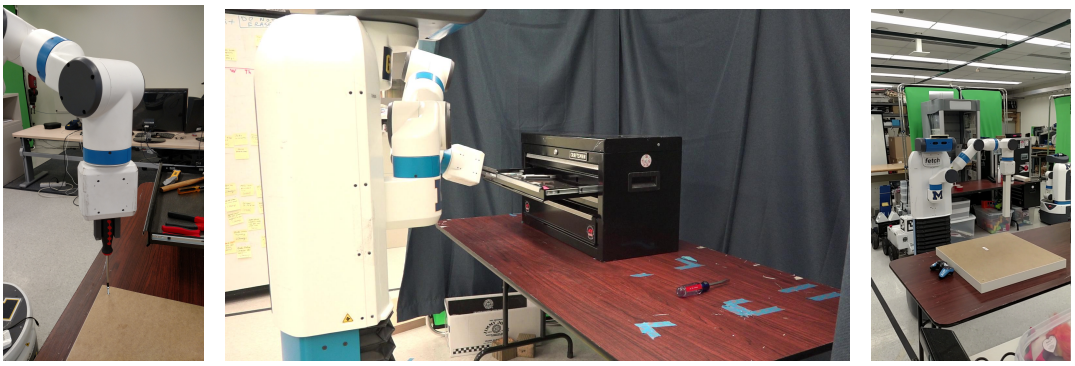
In another project, I studied how different neighborhood selection methods affected PRM constructed. I first evaluated how neighborhood selection methods such as kd-trees and metric trees affected PRT construction time. I then studied how approximate methods such as Spill Trees and DPES affected roadmap quality. This work showed that using approximate neighborhood finders could reduce roadmap construction time significantly for roadmaps with a large number of nodes, and that roadmaps constructed using approximate methods are similar in quality to those constructed using exact methods.

I next studied how introducing randomness to neighbor selection affects roadmap quality [19]. The results of this work indicate that introducing some randomness into neighbor selection resulted in roadmaps that were better connected. As part of this work, I introduced a method called LocalRand (Figure 4) which




(a) (b) (c)

Figure 4: **LocalRand Neighbor Selection: The LocalRand method first locates the k-closest nodes to a sample then selects k of those nodes at random. In this example, LocalRand selects the 5 (green) closest nodes to the sample (red). It then selects 3 (purple) of those nodes at random and attempts to connect them to the sample (green and red edges).**



(a) driving a screw (b) opening a drawer (c) screwing in a table leg

Figure 5: **Affordnaces**

finds a set of nodes that are close to a sample, then selects a subset of these nodes at random. I showed experimentally that LocalRand is able to produce better connected roadmaps than existing methods such as K-Closest, while maintaining a similar cost.

**Motion Planning for Manipulation Affordances:** Affordances provide a natural means for a robot to reason about its environment and the universe of actions available to it [20]. Planning for affordances has applications in areas such as eldercare, manufacturing, and interplanetary exploration.

In order to accommodate affordances, the robot must be able to produce motions that interact with the objects associated with the affordance. These motions must be robust to obstacles in the environment, and they must be able to accommodate any constraints associated with the affordance. To address this problem, I present the concept of affordance wayfields which represent affordances as a cost field over workspace or c-space. I then apply gradient descent planning in wayfields in order to generate motion that enact affordances. Affordance wayfields are advantageous in that they can adapt to obstacles and they can accommodate problem specific constraint. I show that affordance wayfields are able to solve complex tasks such as those shown in Figure 5.

## Future Directions:

I intend to continue working on general motion planning and kinodynamic planning with the objective of further developing tools to better solve motion planning problems and exploring novel applications for motion planning methods. I also plan to continue working on applications of machine learning to motion planning and kinodynamic planning. The next steps in my research will be to further develop the work I am doing on kinodynamic planning for terrains by applying it to more complicated terrains and to different types of robots. I also plan to further develop RVs and to apply them to real-world problems. I will also continue developing methods for applying the concept of affordances to task and motion planning. I also see myself potentially branching out into areas such as belief space planning, parallel robotics, computational geometry, and computational biology.

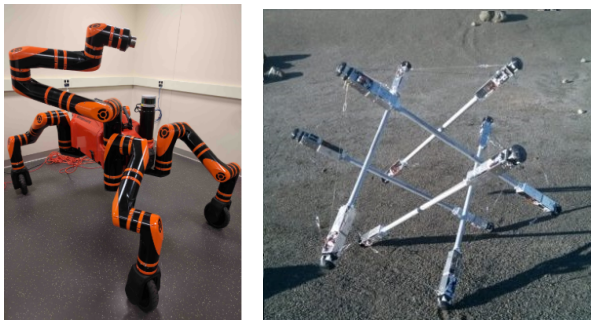
**Expansion of Work in Kinodynamic Planning for Terrains** My next immediate step will be to look at different methods for generating local goals. I am in the process of developing a learned controller that generates local goals for a given terrain. I train this controller using a small number ( 100) of sample paths in the target environment from which I extract tuples of start state, local goal, and global goal. I then use reinforcement learning to train a model that takes as input a state and global goal and returns a local

goal. I will use this model during the expansion step of RRT construction to generate local goals, which I feed to the terrain-aware controller to obtain candidate controls. My expectation is that this will give similar performance to using a wavefront to generate local goals; however, this method will not require the computational cost associated with computing the wavefront,

I am also in the process of developing a version of the terrain-aware planner which only does planning over local terrain. This planner will maintain a local window of terrain around the robot in which it does planning, while applying a high-level representation of the environment (such as a cost-map or low-resolution heightmap) to generate local goals. The main bottleneck with the terrain-aware planner is the cost of simulating large, complex environments, and only needing to simulate a window of the environment will greatly increase the size and resolution of environments the planner can handle. This method will also allow us to apply the terrain-aware planner to real-world applications where the robot only has high-resolution information about the portion of the terrain it can observe with its sensors.

I also plan to extend the terrain-aware controller to more complex terrain environments and robots. In particular, I will develop a variation of the terrain-aware controller that takes into account types of terrain (e.g. pavement, grass, sand, ect.). This controller will improve performance in environments with different types of terrain. I will also develop a terrain-aware controller that can be used by legged robots such as the robomantis (Figure 6(a)). Finally, I plan to apply both reachable volumes and the terrain-aware controller to unique robots such as NASA's tensegrity robot (Figure 6(b)).

**Further Development of Reachable Volumes:** My next objective will be to develop a probabilistic RV, where the RVs of joints are represented by probability distributions. Probabilistic RVs will give us the ability to track the position of joints as probability distributions while enforcing constraints. I foresee them being applied to medical robotics such as steering needles where there is a great deal of uncertainty in the environment and the motions of the robot. I also see them being applied to industrial and home robotics where lower cost, mass produced robots may not be as precise as those encountered in the lab, requiring methods that adapt to noise in robot motion. Probabilistic RVs will allow for planning under uncertainty, they will have applications in belief space planning, and they could be used to apply SLAM-based methods to articulated robots.



(a) (b)

Figure 6: a) Robomantis legged robot b) NASA tensegrity robot

**Applications of Reachable Volumes:** Another area of future research is to explore applying RVs to other problems. I am particularly interested in applying it to computational biology problems such as protein folding. These problems have a large number of degrees of freedom, and are well suited for RVs. To facilitate this I intend to collaborate with bio-medical researchers and with researchers in the drug industry. I also expect that the bio-medical applications of my work will help me to obtain funding from pharmaceutical companies and from organizations such as the NIH.

I would also like to explore applying RVs to folding robots. Motion planning for folding robots has applications in areas such as deformable materials (eg. shape-memory alloys).

I also plan to explore how RVs can improve control and interaction with high degree of freedom robots. This will involve further development of the RV visualization tool so it can be used to provide feedback that will assist in robot control. It will also involve applying reachable volumes and reachable volume sampling

to user guided motion planning where RVs can be used to guide planning and provide feedback.

**Task Planning with Manipulation Affordances:** Affordances provide an atomic representation of actions that could be used for high level task planning. I plan to explore combining the motion planning tools I developed for affordance with high level task planning methods in order to produce complex sequences of motions needed to solve complicated, multi-step tasks. I would also like to incorporate kinodynamics into motion planning for affordances in order to better predict the motion of objects being manipulated. I would like combine the work I am doing with learned goal-reaching controllers with manipulation wayfields by using wayfields to generate local goals which can be used by the controller to generate motions that enact the desired affordance.

**Visual Servoing for Grasping on the Move:** I will develop visual servoing methods for arm control that are suited for grasping on the move. In particular, I will develop visual servoing controls that accommodate the motion of the base and account for the uncertainty that is inherent to grasping on the move. I will also incorporate kinodynamic planning into grasping on the move in order to ensure that trajectories produced by visual servoing are collision-free. I will incorporate these visual servoing methods into the existing framework which I have developed for grasping on the move.

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